ServiceNow Masterclass

Complete Study Notes (Day 1)

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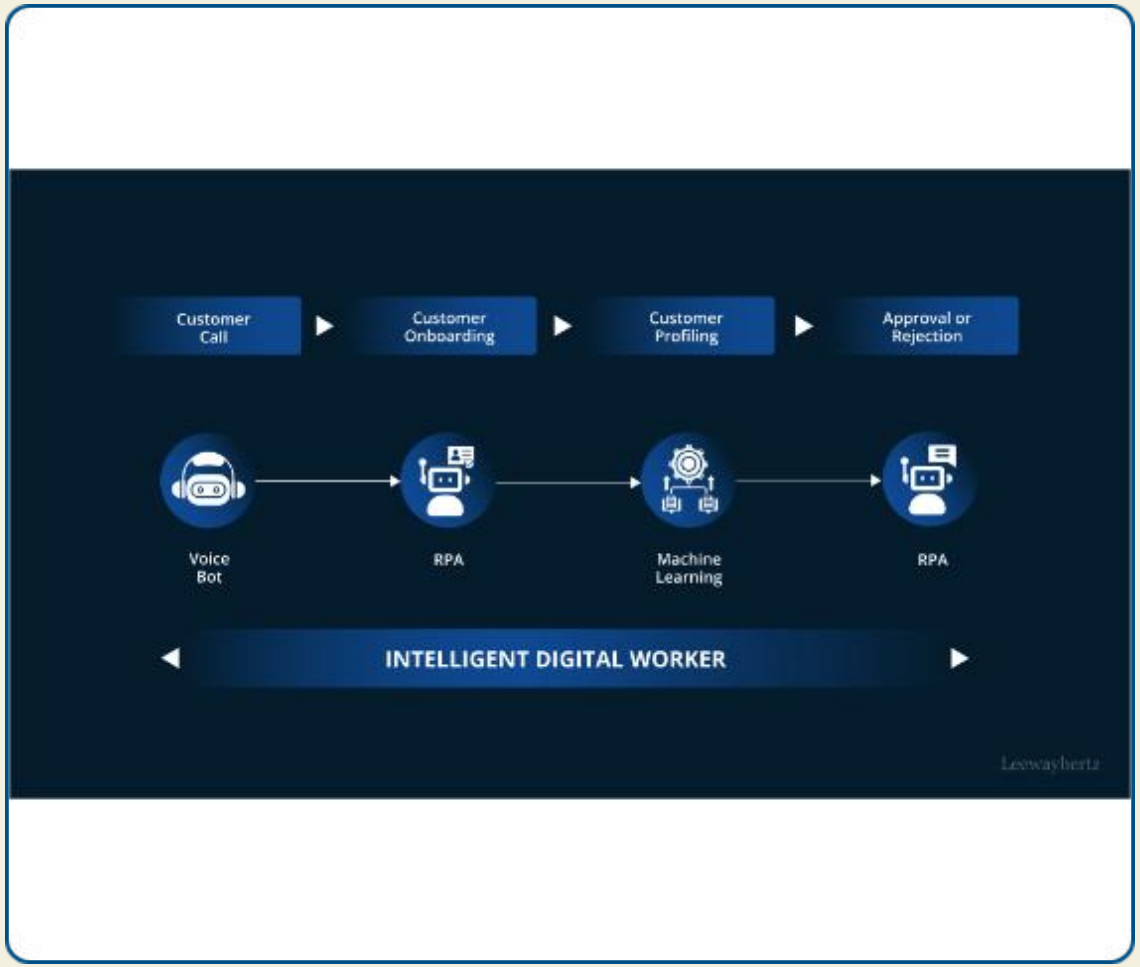
1. What is ServiceNow?

ServiceNow is a cloud-based enterprise platform (SaaS) that enables organizations to automate workflows across IT, HR, customer service, and security.

Key Features

- Cloud platform (no local installation required)
- Low-code / no-code application development
- Centralized, shared data system
- Advanced workflow and process automation



2. Why ServiceNow is Used?

Problems Solved

- **Manual ticket handling:** Replacing emails, phone calls, and spreadsheets.
- **Lack of tracking:** Eliminating the "black hole" where requests get lost.
- **Delayed resolution:** Identifying bottlenecks in business processes.
- **Fragmented systems:** Consolidating disparate tools into one platform.

Benefits Delivered

- **End-to-end Automation:** Routing work instantly to the right team.
- **Transparency:** Real-time dashboards and status updates.
- **Faster Resolution:** Automated responses and self-service portals.
- **Centralized System:** A single source of truth for all enterprise data.

3. ServiceNow Architecture

ServiceNow operates on a highly secure, multi-instance, cloud-hosted architecture built on three fundamental layers.

Presentation Layer

The User Interface (UI). Includes forms, lists, service portals, and dashboards that users interact with daily.

Application Layer

The Logic. Contains Business Rules, Workflows, Client Scripts, and Integrations that process information.

Database Layer

The Storage. MariaDB/MySQL holding all records. **Crucial rule: Everything in ServiceNow is stored in tables.**

4. Core Concepts (Crucial)

1. Table

A collection of data records. Similar to an entire Excel spreadsheet or database table.

2. Record

A single row within a table representing one unique item, user, or incident.

3. Form

A detailed UI view to display, create, or edit a single record from a table.

4. List

A UI grid view displaying multiple records from a table simultaneously.

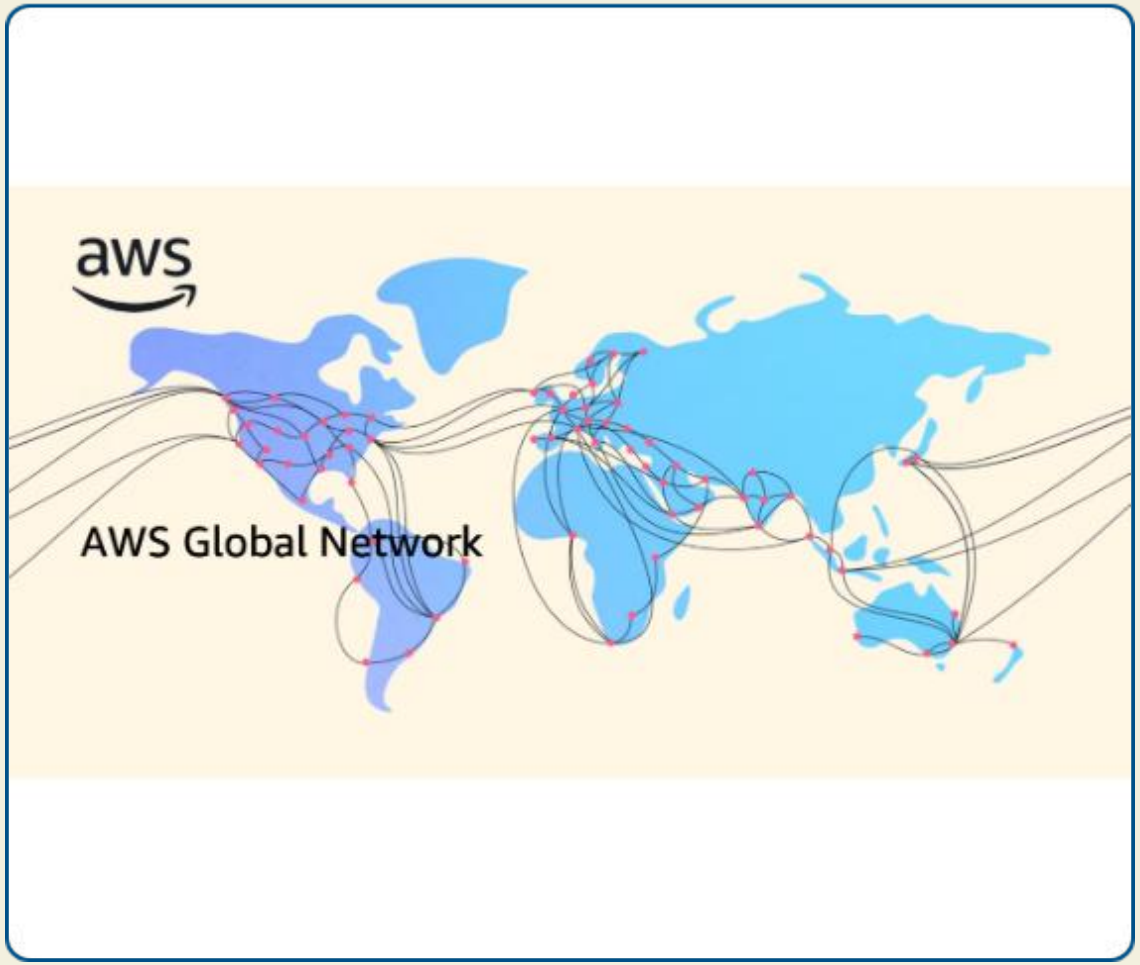
5. ServiceNow Releases

ServiceNow releases major updates twice a year. The naming convention strictly follows an **alphabetical list of major global cities**.

Timeline Examples:

- **T** – Tokyo
- **U** – Utah
- **V** – Vancouver
- **W** – Washington DC
- **X** – Xanadu (Current)
- **Y** – Yokohama (Upcoming)
- **Z** – Zurich
- **A** – Australia (Next Cycle)

** Companies usually stay 1-2 versions behind to ensure stability.*



6. ServiceNow Product Suite

ITSM

IT Service Management (Core)
Incident, Problem, Change, and Request Fulfillment.

ITOM

IT Operations Management
Infrastructure Discovery, Event Management, and Mapping.

HRSD

HR Service Delivery
HR Case management, Onboarding, and Employee Lifecycle.

CSM

Customer Service
Customer support, B2B/B2C portals, and issue resolution.

7. Instance Concept

An instance is an independent, secure environment. Code always flows upwards.

Golden Rule: Never develop directly in Production!

Development (Dev)

Where developers write scripts and build custom applications safely.

Test / QA

Where changes are verified and validated by business stakeholders.

Production (Prod)

The live environment used by real employees and customers.

8. Personal Developer Instance (PDI)

Step 1: Go to the official developer portal at **developer.servicenow.com**.

Step 2: Create an account using your email address and verify it.

Step 3: Click on "Request Instance" and select the latest release version.

Step 4: Save the generated Admin username and password immediately.

Usage: PDIs are free environments used exclusively for practice, learning, and testing ideas without risk.

9. Navigation Overview

Application Navigator

The left-hand sidebar containing all menus, applications, and modules you have access to based on your roles.

Filter Navigator

The search bar at the top of the navigator. **Always use this** instead of manually scrolling to find apps (e.g., type "incident").

Favorites

Click the star icon next to any module to pin it for quick access. Saves significant time for daily, repetitive tasks.

10. Roles and Access Control

ServiceNow uses Role-Based Access Control. Roles define what a user can see and do.

admin

System Administrator. Has absolute, full access to control everything in the platform.

itil

Standard IT user. Can create, edit, and resolve Incidents, Problems, and Changes.

approver_user

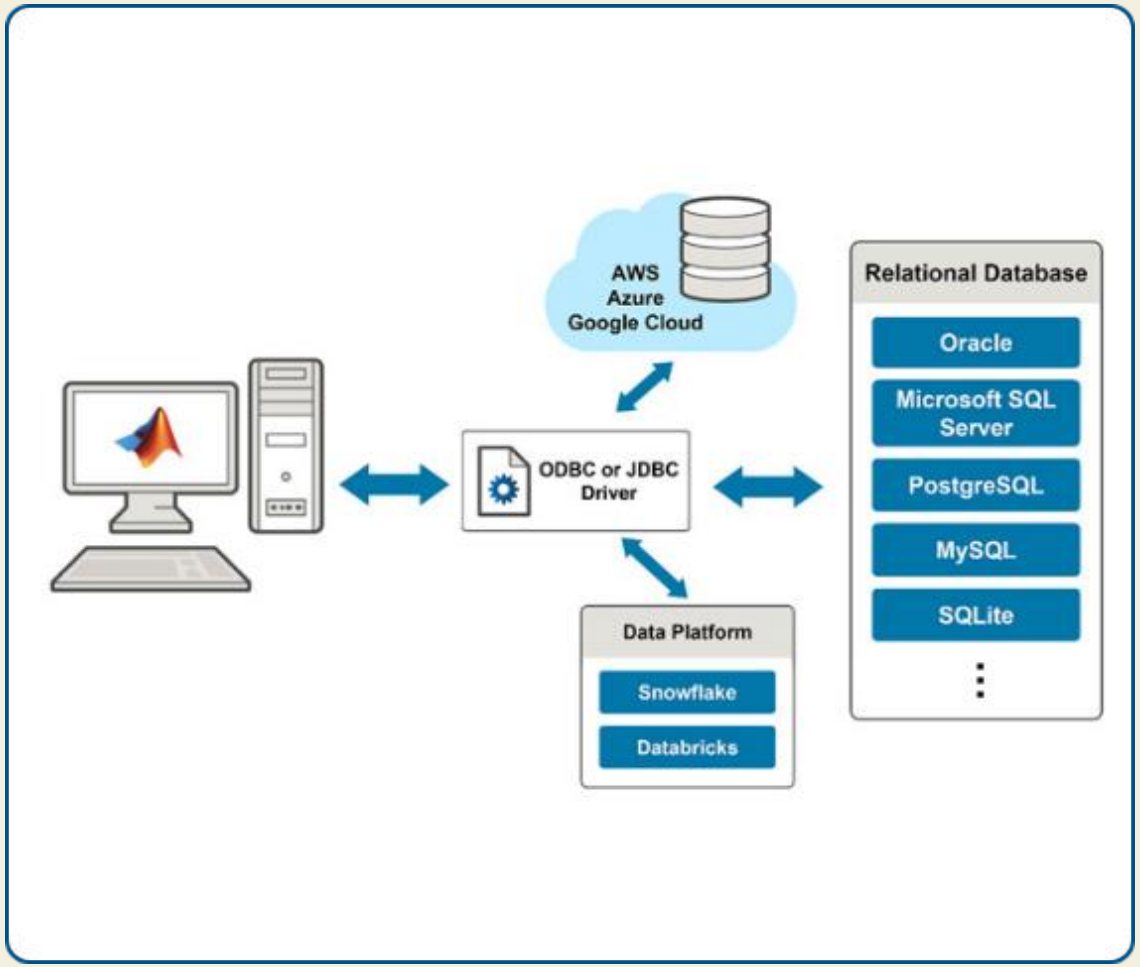
Managers and leads. Allows a user to approve or reject catalog requests and changes.

11. Tables in ServiceNow

Tables store all data across the platform. Every row in the table points to a unique record.

Important System Tables

- **incident** → Stores all Incident tickets.
- **sys_user** → Stores all User profiles.
- **sys_user_group** → Stores all Groups.
- **sys_user_role** → Stores all Roles.
- **task** → The parent table for Incident, Problem, and Change.



12. Forms & Lists Configuration

Form View

Displays a detailed view of a **single record**. It is used to view, enter, or update data for that specific item.

- Example: Viewing the details of Incident INC0010023.
- Admins can configure form layouts to add or hide fields using Form Design.

List View

Displays **multiple records** simultaneously from a table in a grid format.

- Example: Viewing all "Open Incidents" assigned to your group.
- Users can personalize columns to show relevant data using List Layouts.

13. SLA (Service Level Agreement)

SLAs define the agreed-upon time limits for IT to handle an issue, ensuring accountability.

Response Time

15 Min

Time limit to acknowledge and assign the ticket.

Resolution Time

4 Hours

Time limit to completely fix the issue and resolve it.

14 & 15. Workflow & Service Catalog

Workflow / Flow Designer

Modern tools used to automate multi-step business processes without heavy coding.

- Automates Manager **Approvals**.
- Automates **Task Assignments** to groups.
- Sends automated **Email Notifications**.

Service Catalog

The storefront for employees to request hardware, software, or services.

- Creates **Request Items (RITM)** instead of Incidents.
- Example: Requesting a new Laptop or Access to a database.
- Hierarchy: Request (REQ) > Item (RITM) > Task (SCTASK).

16 & 17. Client Script vs. Business Rule

While ServiceNow is low-code, scripting is used for complex logic. Location of execution matters.

Client Script

Runs on the Browser (Frontend)

Executes immediately when a form loads or a user changes a field value.

- Form validation (e.g., proper phone format).
- Showing/Hiding fields dynamically.
- Pop-up alert messages.

Business Rule

Runs on the Server (Backend)

Executes when data is inserted, updated, deleted, or queried in the database.

- Data processing and complex automation.
- Auto-calculating priority before saving.
- Updating related records automatically.

19. Real Use Case: Laptop Issue

1. Create

User raises an Incident ticket.

2. Assign

Business Rule assigns to IT group.

3. SLA Starts

Resolution timer begins tracking.

4. Resolve

IT fixes issue and updates status.

5. Close

Ticket closed. SLA stops. User notified.

20 & 21. Best Practices & Mistakes

Best Practices

- **Practice daily** on your Personal Developer Instance (PDI).
- **Understand basics first.** Master tables and fields before writing any code.
- Use the Filter Navigator instead of manual scrolling.
- Always develop in Dev, never in Production.

Common Mistakes

- **Ignoring fundamentals:** Not understanding how records relate to tables.
- **No hands-on practice:** Watching videos without actually doing the work.
- **Jumping to scripting:** 80% of tasks can be solved with no-code tools.
- **Memorizing instead of understanding** the core concepts.

22. Core Interview Questions

- **What is ServiceNow?**
  A cloud-based (SaaS) workflow automation platform for IT, HR, and business services.
- **What is a table?**
  A collection of data records in the database. (Everything is stored in a table).
- **Difference between Client Script & Business Rule?**
  Client Script runs on the browser (Frontend UI). Business Rule runs on the server (Backend Database).
- **What is SLA?**
  Service Level Agreement. It defines the required response or resolution time for a service.

Final Advice & Tasks

Practice > Theory | Hands-on is key | Build real scenarios

Day 1 Practice Tasks:

1. Request your Personal Developer Instance (PDI).
2. Create 5 new incidents manually.
3. Assign them to specific IT groups.
4. Update their status to resolved.

Image Sources

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Source: www.leewayhertz.com

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